

Round-21: the S-Layer Soundness Kernel in Lean

S4 implies closed frames — the statement WP30/WP32 checked empirically is now a
kernel-checked theorem

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Abstract

The S-layer’s soundness kernel is now in the kernel. `formal/Round19Transport.lean` (2,392 lines, still **zero sorry**) defines the certificate-level S-condition `SCond` — the manuscripts’ S4, stated on the certificate-computable realized network `pairVal` (core net on core pairs, the proved source-oriented update everywhere else) — and proves: a wellformed, S-conditioned certificate unfolds operationally to a frame that is *composition-closed* on every distinct existing triangle (`frame_closed`), *converse-coherent* (`frame_conv`), and *EQ-free* on distinct pairs (`frame_proper`) — i.e. to a proper atomic RCC5 network presentation. This is precisely the claim WP30 Part C and WP32 Part C supported empirically (S4-valid steering $\Rightarrow$ closed unfoldings, 100/100 and 150/150 runs); it is now a theorem, and both poles are witnessed concretely: `certC` satisfies `SCond` and its frame closes end to end, while `certD` — adversarial steering against a `comp(DR, PO)`-pinned geometry, the WP32 Part-C control — provably violates the triangle clause and its frame provably breaks closure at exactly the predicted triple. Status: *strongly supported, not certified*.

1 What was proved

- `uSource_conv`: converse coherence of the source-oriented update (R2). Definitional in every clause except the two crossing a fuel boundary, which the round-20 fuel-irrelevance lemma bridges; the co-birth case is exactly the `net_conv` wellformedness clause round-20 forced.
- `pairVal` and `SCond`: the realized network of a certificate, and the S-condition on it — core net converse-coherent, proper, closed, and agreeing with template core rows (N3); at every step, every triangle through a fresh occurrence composition-closed, and every fresh pair EQ-free. All fields are checkable on a concrete certificate; a `Decidable (p → q)` instance (absent from core Lean) makes the bounded quantifier chains `decide`-friendly.
- `pairVal_closed`: the abstract closure theorem. The proof is a *max-birth rotation argument*: every distinct triple contains an occurrence of maximal birth step; the S-condition covers the triangle with that occurrence in the middle position, and the two kernel-checked rotation facts `comp_rot1/comp_rot2` (finite, by `decide`) together with converse coherence rotate it into the goal orientation. No induction beyond the trichotomy is needed — the birth structure of the fold does all the work.

*Round-21 of the $\mathcal{ALCZ}_{\text{RCC5}}$ decidability project. Substrate: the round-19/20 Lean artifact `formal/Round19Transport.lean` (transport layer defined; characterization and fuel adequacy proved). Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample.

- `frame_closed`, `frame_conv`, `frame_proper`: the operational forms, by transport along round-20's `uSource_eq_frame` plus a new core-row lemma (`get?_corecore`: core-core frame entries carry the core net, because the first write is a pattern write whose core ports the N3 condition pins).
- Witnesses: `certC_scond` with an end-to-end closed-frame example; `certD_wellformed`, `certD_scond_violated` and `certD_frame_not_closed` (values DR/PPI/PO with $\text{DR} \notin \text{comp}(\text{PPI}, \text{PO})$) — the adversarial control of WP32 Part C as kernel-checked theorems.

2 What this closes, and what it does not

Closed: the seam that produced defects #10–#12 (steering conditions versus realized frames) is now theorem-bounded end to end — certificate syntax (`Wellformed`), realized values (`uSource = frame`, round-20), and soundness of the S-condition (closed proper atomic presentations, this round). Not closed: the catalogue level — the manuscripts check S4 on finitely many interface states and conclude closure for *all* unfoldings of the catalogue, while `Cert` encodes one canonical unfolding; lifting `Scond` to a finite catalogue with a reachability fixed point is the natural round-22. Below everything, the standing completeness mathematics (F6 width budgets, W2' uniformization) and the single external input (the RCC5 patchwork property) are untouched. Ledger: thirteen reviews — twelve defects, one no-counterexample; rounds 19–21 unreviewed. The decidability theorem for $\mathcal{ACCT}_{\text{RCC5}}$ remains **strongly supported, not certified**.